EXPERIMENT NO : 3

AIM

To study and observe the operation of basic network devices and packet transmission using the **Cisco Packet Tracer simulation environment**.

PROCEDURE

1. Launch **Cisco Packet Tracer** and open a new workspace.
2. From the **End Devices** section, place four PCs on the workspace.
3. From the **Network Devices → Switches** section, add one switch.
4. Select the **Connections** option and choose a **Copper Straight-Through** cable.
5. Connect each PC to a different Fast Ethernet port of the switch.
6. Verify that the connection indicators become green, showing that the links are active.
7. Open the configuration window of each PC and select **Desktop → IP Configuration**.
8. Assign IP addresses to the four computers in the same network:
    - PC0 – 192.168.10.1
    - PC1 – 192.168.10.2
    - PC2 – 192.168.10.3
    - PC3 – 192.168.10.4
    - Subnet Mask – 255.255.255.0
9. Select the **Add Simple PDU** option (envelope/message icon).
10. Click on a source PC and then select a destination PC to generate a test packet.
11. Change from **Real-Time Mode** to **Simulation Mode**.
12. Start the simulation and observe the movement of the packet from the source computer to the destination through the switch.
13. Examine the events shown in the simulation panel to understand how the switch handles and forwards the packet.

INSTALLATION OF CISCO PACKET TRACER

Step 1: Obtain Cisco Packet Tracer

1. Open the official Cisco learning platform.
2. Sign in using an existing Cisco account or create a free account.
3. Access the Packet Tracer download section available through Cisco's learning resources.

4. Select the version compatible with the operating system being used.

Step 2: Download the Application

1. Choose the required installer for **Windows, Linux, or macOS**.
2. Download the installation package.
3. Save the downloaded file in an easily accessible location.

Step 3: Installation in Windows

1. Locate the downloaded Packet Tracer installer.
2. Double-click the setup file.
3. Accept the Cisco license agreement.
4. Follow the installation wizard by selecting **Next**.
5. Choose the installation directory or retain the default location.
6. Select the required shortcut options.
7. Click **Install** and wait for the process to complete.
8. Select **Finish** after installation.
9. Start Packet Tracer from the Start Menu or desktop shortcut.

Step 4: Installation in Linux

1. Download the appropriate Packet Tracer .deb package.
2. Open the Terminal.
3. Navigate to the directory containing the downloaded package.
4. Install the package using the appropriate apt command.
5. Accept the license agreement when prompted.
6. Complete the installation process.
7. Launch Packet Tracer from the applications menu.

Step 5: Installation in macOS

1. Download the macOS-compatible Packet Tracer installer.
2. Open the downloaded installation file.
3. Follow the instructions displayed by the setup wizard.
4. Accept the license terms.
5. Complete the installation.
6. Open Packet Tracer from the Applications folder.
7. Sign in with the required Cisco learning account if prompted.

NETWORK CONFIGURATION

The following addressing scheme can be used for the experiment:

Device	IP Address	Subnet Mask
PC0	192.168.10.1	255.255.255.0
PC1	192.168.10.2	255.255.255.0
PC2	192.168.10.3	255.255.255.0
PC3	192.168.10.4	255.255.255.0

Since all four PCs belong to the same subnet, they can communicate with each other through the switch without requiring a router.

OBSERVATION

When a packet is generated from one PC to another, the packet first reaches the switch through the connected Ethernet link. The switch examines the destination address and forwards the packet through the appropriate port. In **Simulation Mode**, the complete movement and processing of the packet can be viewed step by step.

RESULT

Thus, the functioning of PCs and a network switch was successfully examined using **Cisco Packet Tracer**. The packet transmission between different computers and the role of the switch in forwarding network traffic were observed successfully.